

SurveilAMR: AI-Enabled Surveillance Intelligence for Global Antimicrobial Resistance Trends

****2026 Vivli AMR Surveillance Data Challenge Submission****

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****Repository:**** [<https://github.com/Nana-Safo-Duker/Vivli-AMR-2026-Data-Challenge-SurveilAMR>]

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Abstract

Industry surveillance data in the Vivli AMR Register remain underused for LMIC decision support. ****SurveilAMR**** provides a reproducible pipeline across four approved datasets—Pfizer ATLAS Antibiotics (1,011,168 isolates; 83 countries; 2004–2024), Paratek KEYSTONE (96,302; 2015–2025), Johnson & Johnson DREAM MDR-TB (5,928; 2011–2019), and Venus Remedies GASAR III (496; India, 2022–2023). We quantified ESKAPE spatiotemporal resistance, beta-lactamase/carbapenemase markers, bedaquiline MICs, and omadacycline potency. *K. pneumoniae* meropenem resistance rose from 4.2% (2007) to 20.7% (2024). African ATLAS isolates (n = 29,387) showed elevated *A. baumannii* meropenem resistance (75.9%). DREAM bedaquiline MICs remained low (median 0.03 µg/mL; Africa 0.06). GASAR revealed ESBL/MBL phenotypes and NDM/VIM co-carriage. Code, tables, and figures are openly available.

Objectives

1. Quantify country/regional AMR trends for empiric therapy and guideline updates.
2. Characterize ESKAPE and priority Gram-negative pathogens against WHO-priority antibiotic classes.
3. Profile resistance mechanisms (CTX-M, KPC, NDM, OXA, SHV, TEM; GASAR gene/phenotype combinations).
4. Detect temporal early-warning signals for antimicrobial stewardship program (ASP) alerts.
5. Integrate ATLAS, KEYSTONE, DREAM, and GASAR for LMIC stewardship, including Ghana Health Service partners.

Methods

****Data sources.**** ATLAS_Antibiotics (Pfizer): metadata, MICs, S/I/R labels, and 23 gene markers. KEYSTONE (Paratek): omadacycline MICs. DREAM (J&J): MDR-TB bedaquiline MICs. GASAR III (Venus Remedies): Gram-negative phenotypic/genotypic combinations and polymyxin B MICs. Aggregates: [repository link](<https://github.com/Nana-Safo-Duker/Vivli-AMR-2026-Data-Challenge-SurveilAMR>).

****Preprocessing.**** Standardized S/I/R labels; minimum ATLAS stratum sizes (n ≥ 20 year–species–drug; n ≥ 50 country–species); parsed MIC inequalities; normalized DREAM TB subtypes and GASAR phenotype buckets.

****Pipeline.**** Python 3.10+ (Pandas, NumPy, SciPy, Matplotlib, Seaborn): chunked ATLAS EDA, resistance estimation, gene prevalence, MDR proxy (≥2 resistant classes), and supplementary summaries (`scripts/run\_analysis.py`, `scripts/analyze\_supplementary.py`, `scripts/generate\_figures.py`). Methods follow 2025 Vivli VIT Grand Prize reproducibility practices.

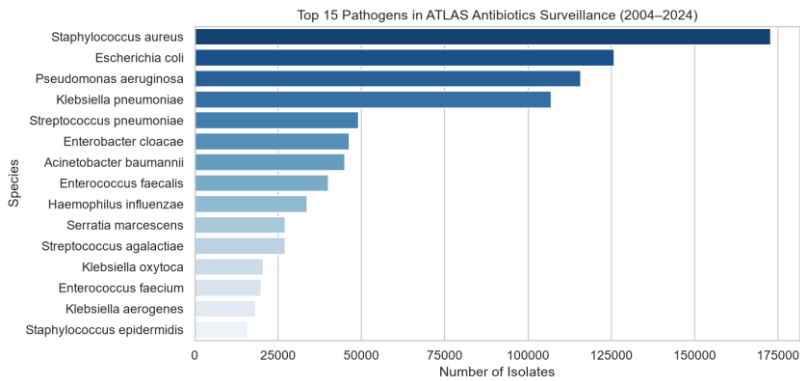
****Limitations.**** Surveillance sampling bias (NA/EU over-representation), incomplete ATLAS gene testing, heterogeneous DREAM subtypes, no patient-level outcomes.

Results

****Table 1. SurveilAMR Vivli Datasets Analyzed****

Dataset	Contributor	Isolates	Years	Geography
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ATLAS Antibiotics	Pfizer	1,011,168	2004–2024	83 countries
KEYSTONE	Paratek	96,302	2015–2025	20 countries
DREAM (MDR-TB)	Johnson & Johnson	5,928	2011–2019	11 countries
GASAR Study III	Venus Remedies	496	2022–2023	India

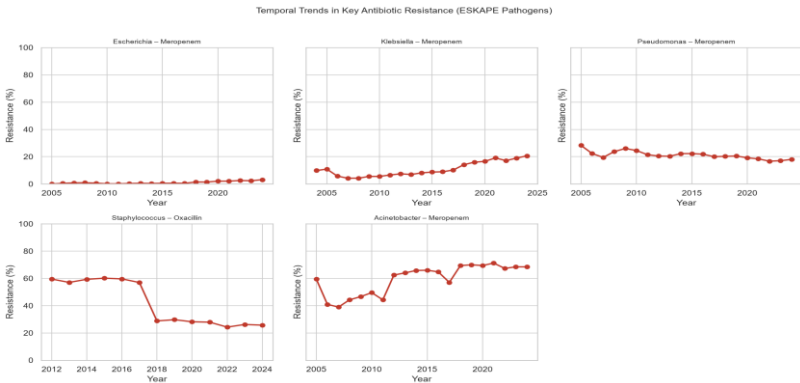
ATLAS dominant species: **S. aureus** (172,883), **E. coli** (125,878), **P. aeruginosa** (115,807), **K. pneumoniae** (106,946). African ATLAS: 29,387 isolates (Ghana 210).



Top 15 pathogens

****Figure 1.**** Top 15 pathogens in ATLAS Antibiotics surveillance (2004–2024).

K. pneumoniae meropenem resistance increased from ****4.2% (2007)**** to ****20.7% (2024)****; **E. coli** to ****3.2% (2024)****; **A. baumannii** remained ****65–71%****; **S. aureus** oxacillin resistance declined to ****25.7% (2024)****.



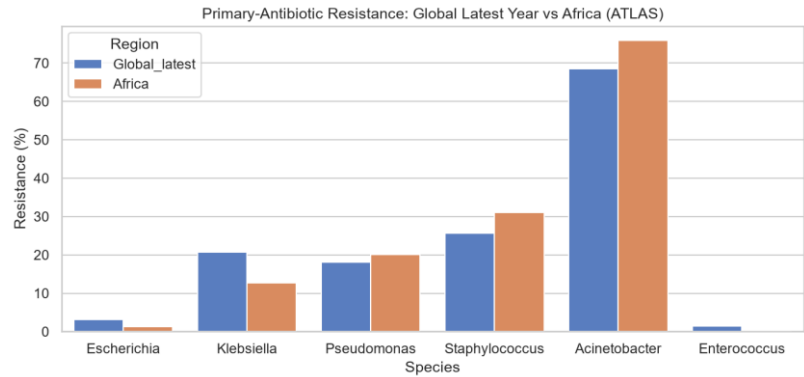
ESKAPE resistance trends

****Figure 2.**** Year-over-year resistance for priority antibiotic–pathogen pairs.

****Table 2. Selected Resistance Rates — Global vs African ATLAS (2024)****

Pathogen	Antibiotic	Global	Africa
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<i>*K. pneumoniae*</i>	Meropenem	20.7%	12.8%

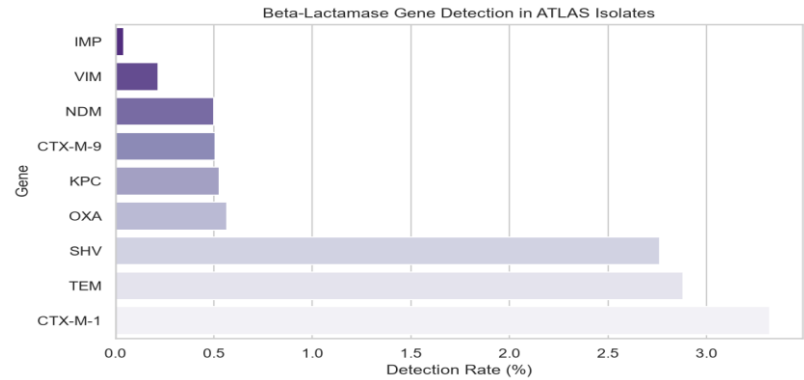
E. coli	Meropenem	3.2%	1.4%
A. baumannii	Meropenem	68.5%	75.9%
S. aureus	Oxacillin	25.7%	31.0%



Global vs Africa resistance comparison

Figure 3. Primary-antibiotic resistance: global latest year vs African ATLAS subset.

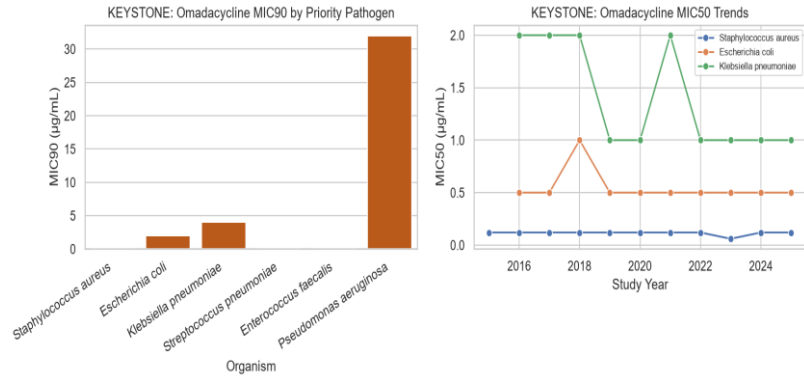
Hotspots included *A. baumannii* meropenem non-susceptibility (Ukraine 93.8%, Jordan 93.0%) and *K. pneumoniae* carbapenem resistance (Ukraine 60.9%, India 53.6%). Top ATLAS genes: CTX-M-1 (3.32%), TEM (2.88%), SHV (2.76%).



Beta-lactamase gene prevalence

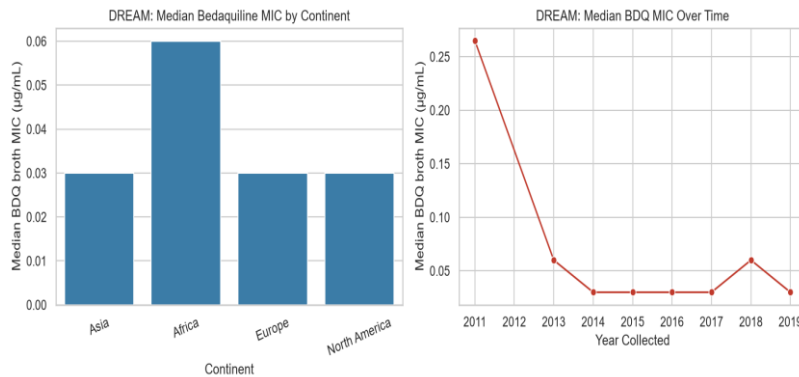
Figure 4. Beta-lactamase and carbapenemase gene detection (ATLAS).

KEYSTONE omadacycline retained potency against *S. aureus* (MIC90 0.25 µg/mL) and *S. pneumoniae* (0.12); DREAM bedaquiline median 0.03 µg/mL (Africa 0.06); GASAR showed ESBL/MBL phenotypes and VIM-1+NDM-1 co-detection (n=41).



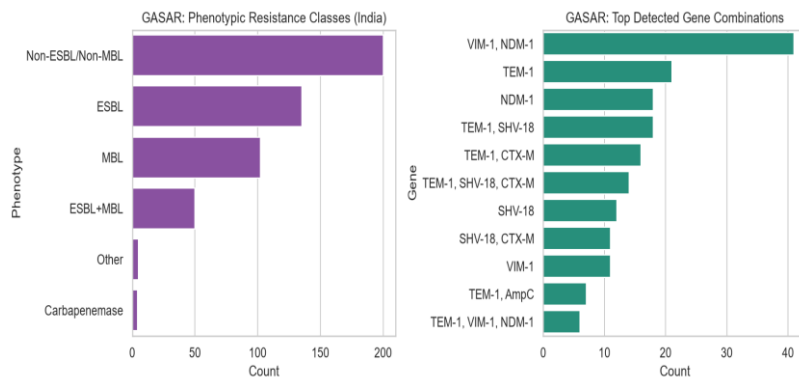
KEYSTONE omadacycline MICs

****Figure 5.** KEYSTONE omadacycline MIC90 by pathogen and MIC50 trends.**



DREAM bedaquiline

****Figure 6.** DREAM bedaquiline MIC distributions by continent and year.**



GASAR mechanisms

****Figure 7.** GASAR phenotypic classes and top gene combinations (India).**

Additional figures (country distribution, heatmaps, MDR proxy, EDA plots) are in `outputs/figures/` and the GitHub repository.

Impact of the Work

SurveilAMR converts restricted Vivli assets into ****open stewardship intelligence**** for Ghana Health Service and the 2026 challenge. Country- and pathogen-specific views support empiric therapy where cultures are delayed. Rising *K. pneumoniae* carbapenem resistance and African *A. baumannii* burden provide ASP triggers; KEYSTONE, DREAM, and GASAR summaries support novel-agent positioning, MDR-TB monitoring, and mechanism-aware prescribing. The [repository](https://github.com/Nana-Safo-Duker/Vivli-AMR-2026-Data-Challenge-SurveilAMR) provides chunked-analysis templates, processed CSVs, and figure scripts following 2025 VIT reproducibility standards. Future work: interactive dashboards and forecasting for African sentinel sites including Ghana.

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